AutoNerve

A unified AI nervous system connecting the supply chain to the shop floor.

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1. What AutoNerve is

AutoNerve is a single AI system that reasons across two signal worlds at once. The **Supply Cortex** watches external signals — news, geopolitics, commodity prices — and decides sourcing. The **Plant Cortex** watches the shop floor — quality, defects, energy — and decides production response. The differentiator is the **bidirectional closed loop** between them: a sourcing decision knows its quality consequence, and a defect on the floor changes the sourcing decision.

Design principle: the LLM decides and explains; deterministic tools compute. A local language model handles extraction and reasoning only. A deterministic engine produces every number — risk, exposure, the sourcing mix — grounded against the knowledge graph so the model cannot hallucinate a decision. Every AI path has a deterministic fallback, so the full system runs and verifies even with no model loaded.

2. Architecture

Layers, from signal to decision:

- **Data / knowledge graph** — the bill-of-materials is the single source of truth: typed entities (products, sub-assemblies, components, raw materials, commodities, suppliers, regions) and typed relationships (contains, made-of, supplied-by, located-in, substitutable-by). Implemented relationally; graph-semantic, traversed by multi-hop reasoning rather than formal inference.
- **OCR intake** — a news-clipping image is read by Tesseract; the extracted text feeds the LLM.
- **LLM layer** — Qwen2.5-1.5B-Instruct runs in-process via HuggingFace transformers (local, no API, no Ollama).
- **Deterministic engine** — graph propagation, MRP exposure, a MILP sourcing optimizer (PuLP/CBC), multi-objective ranking, BOM cost/lead rollup, forecasting, procurement, and the defect loop.
- **Vision** — one-shot reference-vs-test image-diff fault detection (PIL + numpy).
- **Agents** — a chain of role-specialized LLM agents for the end-to-end agentic run.
- **API + UI** — FastAPI serves a single-file command-center dashboard at `/`.

3. The live decision thread (worked example)

The neodymium scenario, traced end to end on the actual data — the demo climax and the strongest single slide:

- **Signal:** a newspaper image — "China widens rare-earth export licensing" — is read by OCR; the LLM extracts the event (material, region, commodity, severity).
- **Graph link:** the material maps to the magnet node; used-in relationships flag two EV models (Sedan + SUV) as exposed.
- **Propagation:** a composite risk score on the China supplier, with attribution from the contributing factors.
- **Exposure:** MRP rolls the disruption to a quantified **Rs 4.2 Cr** at risk over 12 weeks.
- **Optimization:** the MILP solver shifts the magnet mix to Korea + India, cutting China dependency from **100% to 18.5% at +2% cost**, and reports honestly when zero-China is infeasible at full volume without the ferrite substitute.

- **Closed loop:** a vision-QC defect on the China supplier raises its quality-adjusted cost and re-runs the optimizer — the floor changes the sourcing decision.

4. Capability coverage (16 focus areas)

Status reflects the **final build**. **Live** = real working logic in the running system. **Prototyped** = real interface, simplified model. **Roadmap** = named target method, not implemented (and not faked).

#	Capability	Status
1	Risk & bottleneck detection	**Live**
2	Predictive planning (forecast + scenarios)	**Live**
3	Alternate sourcing (MILP)	**Live**
4	Material substitution	**Live**
5	Commodity intelligence (price + scarcity)	**Live**
6	Production-data analysis	**Prototyped**
7	Process capability (Cp/Cpk) + drift forecast	**Prototyped**
8	Low-cost QC automation	**Prototyped**
9	Inventory demand forecasting	**Live**
10	Energy consumption prediction	**Prototyped**
11	Supplier analytics (scorecard)	**Live**
12	ERP-integrated reorder / decision	**Prototyped (mock)**
13	Meeting / action extraction	**Live**
14	Computer-vision defect detection	**Live (one-shot diff)**
15	Operator guidance (SOP retrieval)	**Live**
16	Real-time quality monitoring	**Prototyped**

Coverage note: the supply-chain half (1-5) is fully live — exactly where the brief places its emphasis and its bolded mandate, "reducing geopolitical dependencies."

5. Technology stack

- **Backend:** Python 3.11, FastAPI, PuLP (CBC MILP solver), HuggingFace transformers + PyTorch (Qwen2.5-1.5B, local), Tesseract/pytesseract (OCR), PIL + numpy (vision).
- **Frontend:** a single static HTML file — Tailwind (CDN), globe.gl, inline SVG charts — served by FastAPI. No build step.
- **Data:** CSV/JSON knowledge base (BOM, demand, commodity prices, articles, SOPs, meetings, agent dataset) plus bundled news + part images.

6. Production roadmap (named, not implemented)

The prototype deliberately uses lightweight, defensible methods (deterministic logic + a small grounded LLM + classical CV). The following are the production targets behind the seeded panels, stated as roadmap rather than claimed as built — each defensible in one line, none used as a buzzword in the running system:

- Graph neural network for learned risk propagation (today: deterministic traversal + attribution).
- Temporal transformer / NHITS / DeepAR for demand (today: least-squares trend with risk-adjusted bands).
- YOLOv8 + PatchCore for un-referenced defect detection (today: one-shot reference-vs-test diff).
- Whisper for meeting-audio intake (today: schema-bound extraction from transcript text).
- Altman-Z financial-health model for supplier scoring (today: a transparent composite).
- Streaming + time-series infrastructure (Kafka / time-series DB / object store) for live multi-plant deployment.

7. What is real vs modeled (honesty statement)

- **Genuinely real and running:** OCR (Tesseract), local LLM extraction (Qwen), the MILP sourcing optimizer, knowledge-graph propagation, demand forecasting, one-shot vision diff, and the defect-to-re-sourcing closed loop.
- **Modeled / illustrative:** all demo data is synthetic; savings and risk-score figures are modeled on stated assumptions; some breadth panels (energy, ERP push) are seeded displays.
- **Scope discipline:** sophisticated methods that are not implemented are labelled roadmap, never presented as running. This makes every claim defensible under technical questioning.

8. Scalability

The graph + optimizer architecture scales by adding rows, not rewrites. Today: one plant, five products, twenty suppliers. The same design extends to many plants, hundreds of products, and thousands of suppliers, with the streaming/time-series infrastructure above added underneath for live signal ingestion.